

Atmospheric Technosignatures in ExoFarm Scenarios around M-Dwarf Stars

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Background

Nitrogen is essential to life, and industrial agriculture substantially alters its planetary cycle through fertilizer use and nitrogen fixation. On Earth, agriculture accounts for approximately 65% of global NH_3 emissions and 26–30% of N_2O emissions (Haqq-Misra et al., 2022; Tian et al., 2024). As food production may remain a sustained technological activity, these perturbations motivate ExoFarm as a potential long-term atmospheric technosignature (Haqq-Misra et al., 2022). However, atmospheric abundances depend not only on surface emissions but also on photochemistry shaped by stellar irradiation and chemical sinks (Segura et al., 2005; Huang et al., 2022). Understanding this connection is essential to assessing whether agricultural signatures could be identified in exoplanet spectra.

Question

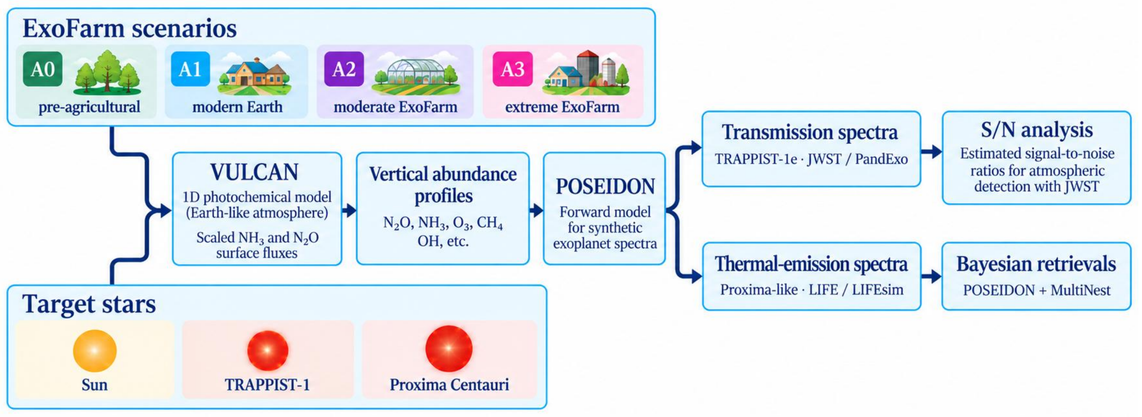
How does stellar irradiation affect the atmospheric accumulation of N_2O and NH_3 from large-scale agriculture, and how well could these gases be detected and retrieved in exoplanet spectra?

Methods

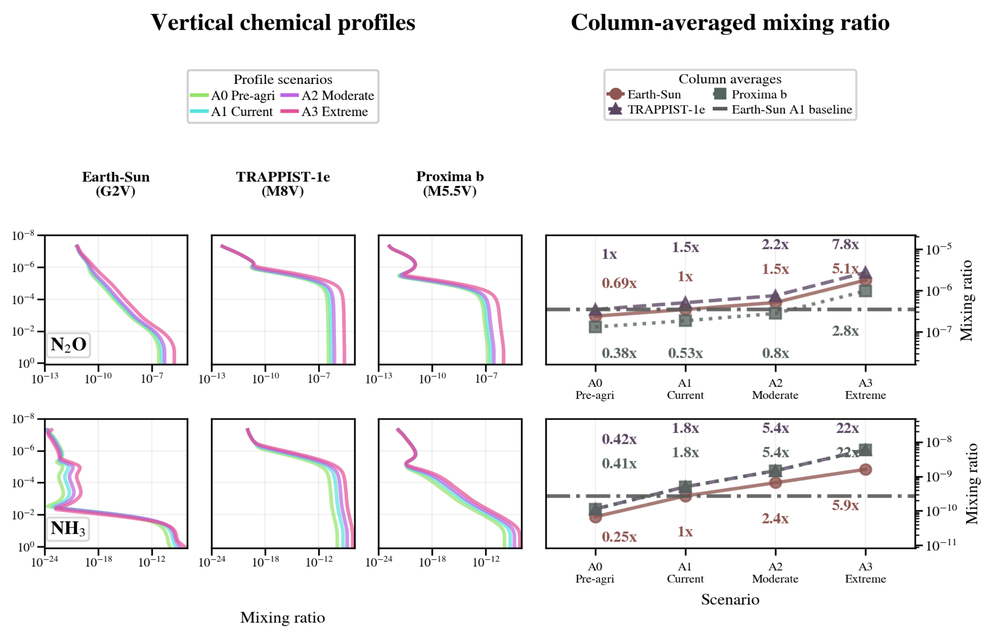
Rather than prescribing atmospheric abundances, we scaled the surface fluxes of N_2O and NH_3 relative to the pre-agricultural (A0) and modern-Earth (A1) benchmarks:

$$F_i(A_j) = F_i(A0) + \alpha_{i,j}[F_i(A1) - F_i(A0)]$$

The moderate scenario (A2) uses $(\alpha_{NH_3}, \alpha_{N_2O}) = (3.50, 2.55)$, while the extreme scenario (A3) uses (15, 15), following Haqq-Misra et al. (2022, 2025).
We used VULCAN to compute steady-state atmospheric compositions with prescribed P–T and Kzz profiles. Synthetic spectra were generated with POSEIDON, with instrument sensitivity evaluated using PandExo and LIFEsim. Bayesian retrievals of the LIFE simulations were performed with POSEIDON and MultiNest.



Photochemical Response



We modeled the A0–A3 scenarios using VULCAN's Earth-validated S–N–C–H–O photochemical network (Tsai et al., 2017, 2021) under irradiation from the Sun, TRAPPIST-1, and Proxima Centauri. Apart from the scaled N_2O and NH_3 surface fluxes, the remaining chemical lower-boundary conditions were retained from the modern-Earth configuration.
The atmospheric response depends strongly on the host star. Although N_2O follows changes in surface emissions approximately proportionally within each environment, its abundance and vertical distribution differ across stellar conditions. NH_3 shows an even stronger dependence on atmospheric processing, with substantially greater accumulation and vertical persistence under TRAPPIST-1 irradiation.

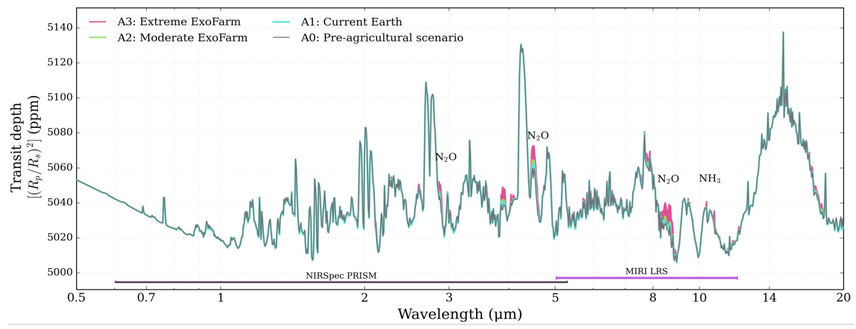
Spectral Signatures and Detectability

Using atmospheric profiles computed with VULCAN, we generated transmission spectra for a TRAPPIST-1e analog and thermal-emission spectra for an Earth-like Proxima b analog with POSEIDON. To quantify the spectral contribution of agricultural N_2O and NH_3 enhancement, we compared each spectrum with one in which the selected molecular profile was restored to its pre-agricultural (A0) state. Molecular sensitivity was then evaluated across the instrumental wavelength channels using uncertainties from PandExo and LIFEsim.

$$S/N_{X,k} = \left[\sum_{i \in k}\left(\frac{\Delta_{X,i}}{\sigma_i}\right)^2\right]^{1/2}$$

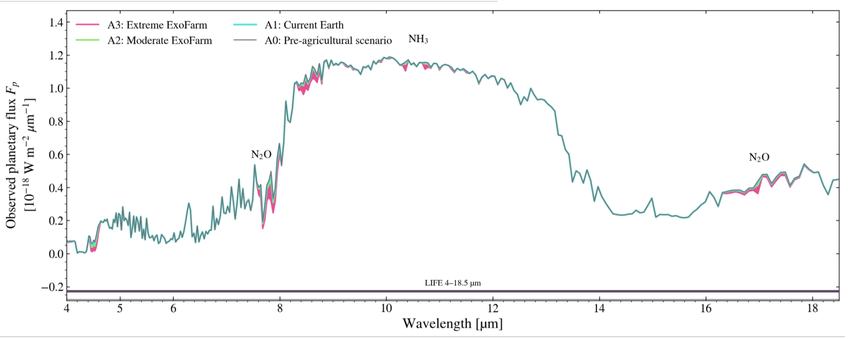
Although the A0–A3 spectra largely overlap, their molecular contributions reveal markedly different observational requirements. The modeled LIFE configuration offers greater sensitivity to N_2O than the JWST transmission case, while NH_3 remains considerably more challenging. These integrated S/N estimates assume known molecular signals; Bayesian retrievals are needed to evaluate their identification amid atmospheric degeneracies.

Transmission spectrum
Trappist-1e analog



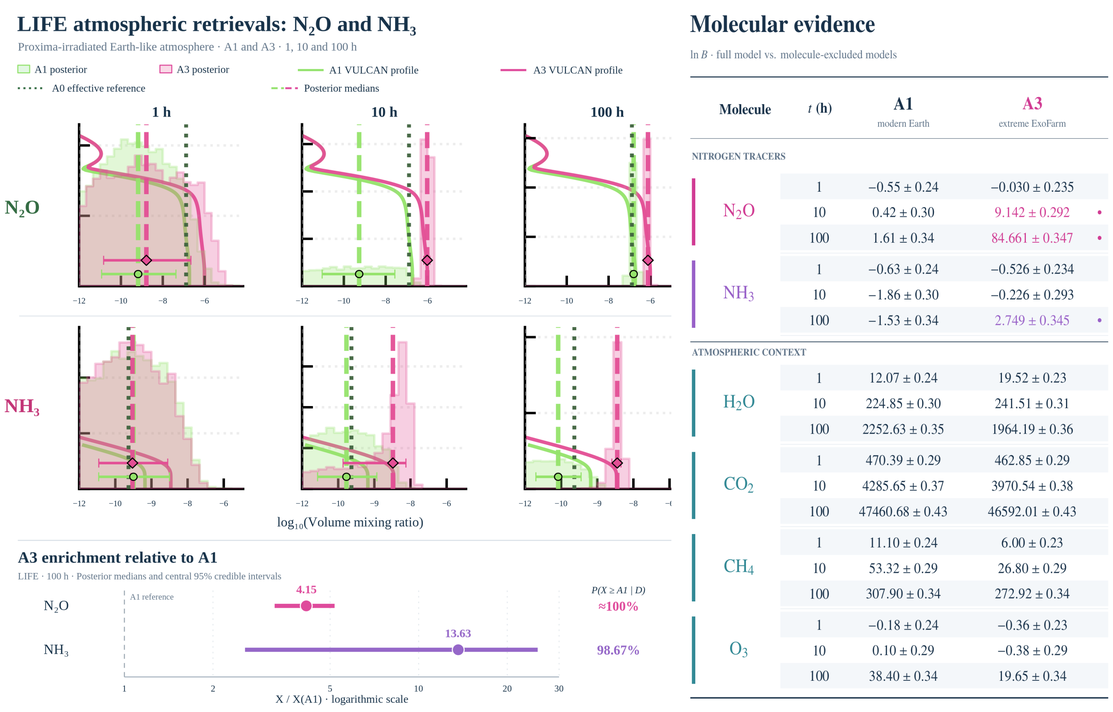
JWST · Transmission
TRAPPIST-1e · A3
2.9
Integrated N_2O S/N
100 NIRSpec PRISM transits
NH_3 remains below S/N = 1 after 100 transits.
Projected N_2O S/N = 5: ~290 transits

Thermal emission spectrum
Proxima b analog



LIFE · Thermal Emission
Proxima b analog · A3 · 10 h
5.97 N_2O S/N | **1.54** NH_3 S/N
Projected time to reach S/N = 5
N_2O ~7 h · NH_3 ~106 h

Bayesian Retrievals and ExoFarm Enrichment



Molecular evidence

ln B - full model vs. molecule-excluded models

Molecule	t (h)	A1 modern Earth	A3 extreme ExoFarm	
NITROGEN TRACERS				
N_2O	1	−0.55 ± 0.24	−0.030 ± 0.235	
	10	0.42 ± 0.30	9.142 ± 0.292	•
	100	1.61 ± 0.34	84.661 ± 0.347	•
NH_3	1	−0.63 ± 0.24	−0.526 ± 0.234	
	10	−1.86 ± 0.30	−0.226 ± 0.293	
	100	−1.53 ± 0.34	2.749 ± 0.345	•
ATMOSPHERIC CONTEXT				
H_2O	1	12.07 ± 0.24	19.52 ± 0.23	
	10	224.85 ± 0.30	241.51 ± 0.31	
	100	2252.63 ± 0.35	1964.19 ± 0.36	
CO_2	1	470.39 ± 0.29	462.85 ± 0.29	
	10	4285.65 ± 0.37	3970.54 ± 0.38	
	100	47460.68 ± 0.43	46592.01 ± 0.43	
CH_4	1	11.10 ± 0.24	6.00 ± 0.23	
	10	53.32 ± 0.29	26.80 ± 0.29	
	100	307.90 ± 0.34	272.92 ± 0.34	
O_3	1	−0.18 ± 0.24	−0.36 ± 0.23	
	10	0.10 ± 0.29	−0.38 ± 0.29	
	100	38.40 ± 0.34	19.65 ± 0.34	

We performed Bayesian retrievals of the A1 and A3 LIFE spectra (1, 10, and 100 h) using POSEIDON/MultiNest, fitting isochemical abundances of H_2O, CO_2, CH_4, O_2, O_3, N_2O, and NH_3, a six-parameter Madhusudhan–Seager (2009) pressure–temperature profile, and planetary radius. Since the synthetic spectra preserve VULCAN's vertical chemistry, the retrievals constrain effective atmospheric abundances. Molecular evidence was evaluated by comparing the full model with molecule-excluded models, allowing the remaining parameters to readjust.
H_2O, CO_2, and CH_4 receive strong evidence even at 1 h, while O_3 requires longer exposures. In A3, N_2O is strongly supported by 10 h, whereas NH_3 provides moderate evidence at 100 h; both remain weakly supported in A1. Beyond molecular identification, we evaluated individual and joint enrichment relative to A1, comparing posterior probabilities with prior odds. At 100 h, N_2O is relatively well constrained, while NH_3 shows a larger but less precise enrichment. Their joint recovery above A1 reaches 98.67% posterior probability, with a 114-fold increase in odds, supporting an intensified nitrogen cycle without uniquely establishing an agricultural origin.

Take-Home Message

Under the stellar spectra explored, M-dwarf irradiation favors the accumulation and vertical persistence of N_2O and NH_3, suggesting that strong agricultural forcing could sustain enhanced NH_3 even in oxidizing, Earth-like atmospheres. While these perturbations remain challenging to distinguish with JWST transmission spectroscopy, LIFE may extend its potential for atmospheric characterization and biosignature searches to the investigation of agricultural technosignatures.
However, molecular S/N provides only an initial sensitivity estimate; Bayesian retrievals are needed to assess molecular evidence and atmospheric degeneracies. LIFE simulations strongly support H_2O, CO_2, and CH_4 even at short exposures, with O_3 becoming accessible at longer integrations. In contrast, the nitrogen tracers become informative mainly in the extreme ExoFarm scenario: N_2O receives strong evidence by 10 h, while NH_3 provides moderate evidence at 100 h. At this exposure, their joint enrichment above A1 is strongly supported, increasing the odds of an intensified nitrogen-cycle state by approximately 114 times relative to the prior.

Limitations & Future Work

ExoFarm scenarios prescribe scaled N_2O and NH_3 surface fluxes without a biogeochemical model linking agricultural intensity to emissions. The simulations also assume fixed atmospheric thermal structures and vertical mixing, cloud-free spectra, and simplified instrumental noise. Photochemical predictions remain sensitive to the adopted stellar UV spectra and their spectral discretization (Sneed et al., 2026).
Moreover, large N_2O abundances may arise naturally under different planetary conditions (Schwieterman et al., 2022), while NH_3 accumulation depends strongly on surface sinks (Huang et al., 2022). Future work should address these uncertainties, stellar variability, and atmospheric feedbacks, ultimately moving toward retrievals of surface emission fluxes through coupled photochemical and biogeochemical models, rather than relying solely on isochemical abundances.

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